CSE486 - Big-Data Analytics

CARS PRICES DATASET

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PHASE 1: DISCOVERY

1.1 Dataset Overview

The dataset contains 19,237 car listings with 18 attributes. Key features include numeric variables such as Price, Levy, Engine Volume, Mileage, and categorical variables including Manufacturer, Model, Fuel Type, Gear Box Type, and Category.

Table 1: Key Features of the Dataset

Feature	Type	Description
Price	Numeric	Car selling price (target variable)
Levy	Numeric	Tax/fee associated with the car
Engine_volume	Numeric	Engine displacement in liters
Mileage	Numeric	Distance travelled by the car (km)
Manufacturer	Categorical	Car manufacturer
Model	Categorical	Car model
Category	Categorical	Vehicle category (SUV, Sedan, etc.)
Fuel Type	Categorical	Type of fuel used
Gear Box Type	Categorical	Manual, Automatic, or other

1.2 Problem Statement

- Business Problem: Accurately predict car prices to support dealerships, buyers, and insurance assessments.
- Data Science Problem: Regression task predicting continuous Price from numeric and categorical features.

1.3 Objectives

1. Clean and preprocess the dataset for modeling.
2. Detect and handle outliers and missing values.
3. Explore relationships between car features and price.

4. Build predictive models using Linear Regression, Decision Trees, and Random Forest.
5. Evaluate models using RMSE and R^2.
6. Test hypotheses related to vehicle features and price.

1.4 Hypotheses

- H1: Automatic vs Manual → t-test.
- H2: Fuel type → ANOVA + Tukey HSD.
- H3: Leather interior → t-test.
- H4: Mileage correlation → Spearman correlation.

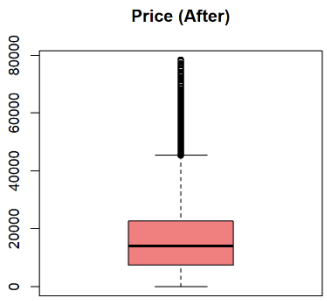
PHASE 2: DATA PREPARATION

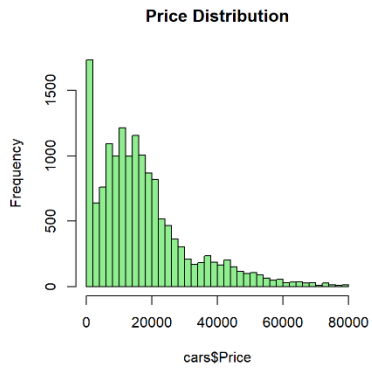
2.1 Data Loading and Cleaning

- Dataset loaded using read_csv() with missing entries "-" replaced by NA.
- Removed 3,512 duplicate rows.
- Column types converted: numeric (Levy, Engine_volume, Mileage_km) and categorical (Manufacturer, Model, Category, etc.).

2.2 Outlier Detection

- Applied IQR method (multiplier = 3.5) to detect outliers in Price, Mileage_km, and Engine_volume.
- Outliers removed to prevent skewed model results.





2.3 Missing Value Imputation

- Manufacturers with >30% missing Levy treated as structural missing so Levy set to 0.
- Remaining missing Levy values imputed with the median because our dataset is skewed.

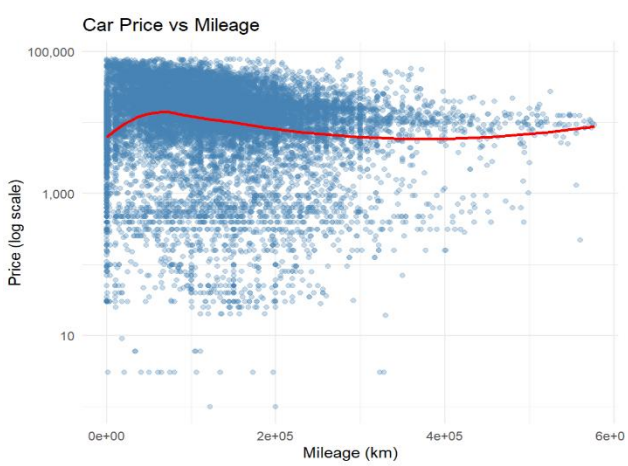
2.4 Train-Test Split

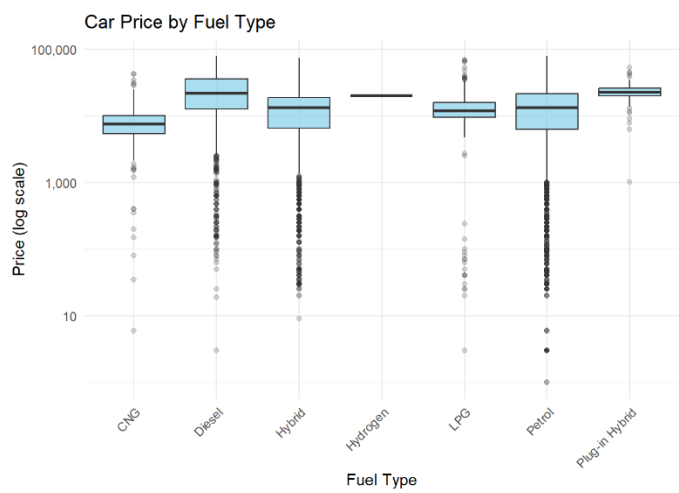
- Random 80-20 split for training and testing sets.
- Low-frequency manufacturers grouped into "Other" to handle unseen levels in the test set.

PHASE 3: MODEL PLANNING

3.1 Exploratory Data Analysis (EDA)

- Price vs Mileage: Negative correlation observed (higher mileage → lower price).
- Price vs Production Year: Positive correlation (newer cars → higher price).
- Price by Fuel Type & Category: Boxplots show significant differences.





3.2 Feature Selection

Selected variables:
Price, Prod.year, Levy, Engine_volume, Turbo, Mileage_km, Cylinders, Doors, Airbags, Manufacturer, Category, Fuel type, Gear box type, Drive wheels, Wheel.

3.3 Analytics Technique Selection

- Linear Regression: Baseline continuous prediction model.
- Decision Tree: Captures nonlinear relationships and interactions.
- Random Forest: Robust to outliers, handles categorical variables, provides feature importance.
- K-Means Clustering: For grouping similar cars.

PHASE 4: MODEL BUILDING

4.1 Model Training

- Linear Regression: log-transformed Price to normalize skewed distribution.
- Decision Tree: Trained using rpart(), complexity controlled by cp = 0.01.
- Random Forest: 300 trees (ntree = 300), 4 variables per split (mtry = 4).

4.2 Model Evaluation

Three predictive models were trained to estimate car prices: **Linear Regression**, **Decision Tree**, and **Random Forest**. The target variable was log-transformed to reduce skewness in the price distribution and improve model performance.

Evaluation metrics (on the 20% held-out test set) include **Root Mean Squared Error (RMSE)** and **R-squared (R^2)**:

Model	RMSE	R-squared
Linear Regression	1.25	0.217
Decision Tree	1.17	0.311
Random Forest	1.01	0.484

Interpretation:

- The **Random Forest** model outperformed both Linear Regression and Decision Tree models, achieving the lowest RMSE (1.01) and the highest R^2 (0.484). This indicates it explains ~48% of the variance in log-transformed car prices.
- The **Decision Tree** performed moderately well, with better fit than Linear Regression but lower predictive accuracy than Random Forest.

- The **Linear Regression** model had the highest RMSE and lowest R^2, suggesting that linear assumptions alone are insufficient to capture the complex relationships between car features and price.

Feature Importance (Random Forest)
Random Forest analysis revealed the most influential features for predicting car prices:

Rank	Feature	% Increase in MSE
1	Airbags	81.10
2	Gear box type	72.85
3	Production Year	70.18
4	Manufacturer	43.33
5	Levy	43.16
6	Mileage (km)	42.35
7	Fuel Type	37.27
8	Engine Volume	35.18
9	Category	34.56
10	Turbo	33.36

Interpretation:

- Features like **Airbags**, **Gear box type**, and **Production Year** are the strongest predictors of car price.
- Both mechanical characteristics (engine volume, turbo, gear type) and structural features (manufacturer, airbags, fuel type) play significant roles in price determination.

4.3 Hypothesis Testing Results

H1: Effect of Transmission Type on Car Prices

To examine whether cars with automatic transmissions are more expensive than manual ones, a **Welch two-sample t-test** was conducted. The results indicate a statistically significant difference between the two groups:

- Mean price for automatic cars: **$16,926**
- Mean price for manual cars: **$11,385**
- $t = 20.26$, $df = 3419.4$, **p-value < 0.001**
- 95% confidence interval for the mean difference: [$5,004, $6,077]

Interpretation: Automatic cars are significantly more expensive than manual cars, confirming the hypothesis (H1).

H2: Effect of Fuel Type on Car Prices

An **ANOVA test** was conducted to determine whether the type of fuel affects car prices. The analysis yielded:

- $F(6, 15222) = 225.1$, **p-value < 0.001**

Interpretation: The type of fuel significantly impacts car prices. A post-hoc **Tukey HSD test** was attempted to compare all pairs of fuel types; however, a minor error occurred in generating the comparisons. Nonetheless, the ANOVA results alone strongly indicate a significant effect of fuel type on car price, supporting hypothesis H2.

H3: Effect of Leather Interior on Car Price

We tested whether cars with leather interiors have significantly different prices compared to cars without leather interiors. Using a Welch Two-Sample t-test on 14,819 observations, the results indicate a highly significant difference in mean prices between the two groups:

- **t-statistic:** -28.784
- **Degrees of freedom:** 14,818
- **p-value:** < 0.0001
- **95% Confidence Interval:** [-6266.37, -5467.34]
- **Mean Price:**
    - Cars without leather interior: 13,308.87
    - Cars with leather interior: 19,175.72

Interpretation: Cars with leather interiors are on average approximately 6,000 higher in price than cars without leather interiors. The p-value confirms that this difference is statistically significant, supporting the hypothesis that leather interior affects car price.

H4: Correlation between Mileage and Price

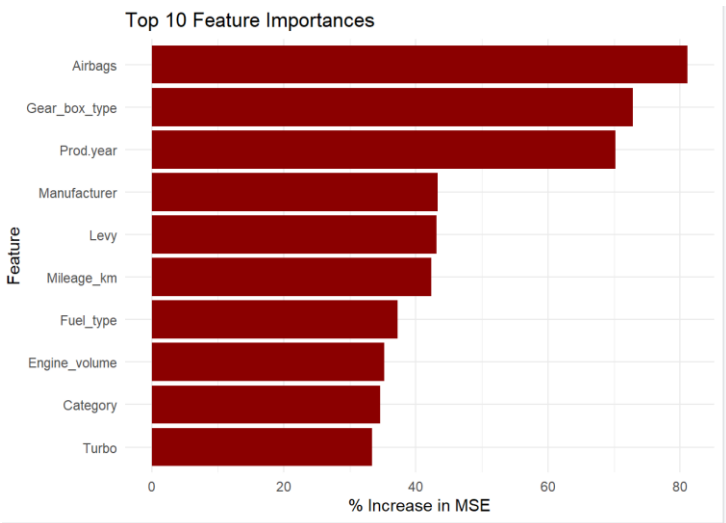
To investigate whether car mileage is associated with price, we calculated Spearman's rank correlation coefficient due to potential non-linear relationships and the presence of outliers.

- **Spearman's rho:** -0.170
- **p-value:** < 0.0001

Interpretation: There is a weak but statistically significant negative correlation between mileage and price, indicating that cars with higher mileage tend to have slightly lower prices. The correlation is significant despite being weak due to the large sample size (n = 15,725).

PHASE 5: COMMUNICATE RESULTS

5.1 Feature Importance (Random Forest)



This bar chart illustrates the relative importance of each feature in the Random Forest model, measured by the percentage increase in Mean Squared Error (%IncMSE) when the feature's values are permuted.

Features such as **Airbags**, **Gear Box Type**, and **Production Year** show the highest importance scores, indicating that changes in these variables lead to the largest decrease in prediction accuracy. This means they play a critical role in determining car prices.

Lower-ranked features, such as **Turbo** and **Category**, still contribute to the model but have a smaller individual impact. Overall, the plot confirms that both **vehicle specifications** and **categorical attributes** significantly influence car price predictions.

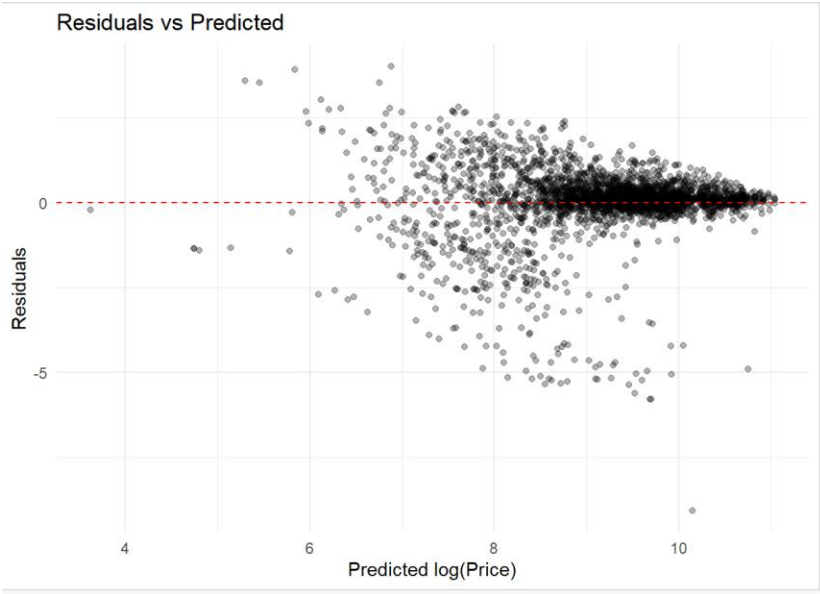
5.2 Visualizations

Since the Random Forest model outperformed both the Linear Regression and Decision Tree models in terms of RMSE and R-squared, the following plots present its evaluation results:

This scatter plot compares the actual car prices with the prices predicted by the model. Each point represents a single observation, where perfect predictions would lie exactly on the 45-degree diagonal line.

The points are generally clustered around the diagonal, indicating that the model is able to capture the overall relationship between the input features and car prices. However, some dispersion is visible, especially at higher price values, suggesting that the model's predictions become less accurate for very expensive vehicles.

Overall, this visualization confirms that the model performs reasonably well, while highlighting areas where prediction errors increase.



This plot shows the residuals (actual price − predicted price) plotted against the predicted prices. Ideally, residuals should be randomly scattered around zero with no clear pattern.

The residuals in this plot are centered around zero, indicating that the model does not systematically overestimate or underestimate car prices. However, the spread of residuals increases slightly for higher predicted prices, suggesting **heteroscedasticity**, where prediction errors grow as price increases.

Despite this, the absence of strong patterns or curves implies that the model is appropriately specified and that major nonlinear relationships have been captured.